

# Project Planning Phase

## Project Planning

|               |                       |
|---------------|-----------------------|
| Date          | 30 June 2026          |
| Roll Number   | 23A51A4294            |
| Project Name  | Leave Approval System |
| Maximum Marks | 5 Marks               |

### Product Backlog, Sprint Schedule, and Estimation

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                   | Story Points | Priority | Team Members           |
|----------|-------------------------------|-------------------|-----------------------------------------------------------------------------------------------------|--------------|----------|------------------------|
| Sprint-1 | Leave Application             | USN-1             | As an employee, I can apply for leave by selecting leave type, start date, and end date.            | 3            | High     | Developer-1, Tester-1  |
| Sprint-1 | Validation                    | USN-2             | As an employee, I want the system to validate that the end date is not earlier than the start date. | 2            | High     | Developer-2            |
| Sprint-2 | Manager Approval              | USN-3             | As a manager, I can review and approve or reject leave requests through Flow Designer.              | 3            | High     | Developer-1, Manager-1 |
| Sprint-2 | Notifications                 | USN-4             | As an employee, I will receive an email or SMS notification once my leave is approved or rejected.  | 2            | Medium   | Developer-2            |
| Sprint-3 | Leave Records                 | USN-5             | As an HR admin, I can view all leave requests and approval history in the dashboard.                | 3            | Medium   | Developer-1, Tester-2  |
| Sprint-3 | Reporting                     | USN-6             | As an admin, I can generate monthly leave reports for all employees.                                | 2            | Low      | Developer-2            |
| Sprint-4 | Integration                   | USN-7             | Integrate the system with HR database and ServiceNow APIs for centralized data management.          | 3            | Medium   | Developer-1, Tester-1  |

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                              | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|--------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-4 | UI Enhancement                | USN-8             | Improve the user interface for better accessibility and mobile responsiveness. | 2            | Low      | Developer-2  |

### Project Tracker, Velocity & Burndown Chart

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-1 | 10                 | 6 Days   | 01 July 2026      | 06 July 2026              | 10                                              | 06 July 2026                 |
| Sprint-2 | 10                 | 6 Days   | 08 July 2026      | 13 July 2026              | 9                                               | 13 July 2026                 |
| Sprint-3 | 10                 | 6 Days   | 15 July 2026      | 20 July 2026              | 10                                              | 20 July 2026                 |
| Sprint-4 | 10                 | 6 Days   | 22 July 2026      | 27 July 2026              | 9                                               | 27 July 2026                 |

### Velocity Calculation

If the team completes 20 story points in a 10-day sprint,

$$\text{Average Velocity (AV)} = \frac{20}{10} = 2 \text{ story points per day.}$$

### Burndown Chart

The burndown chart will show a steady decline in remaining story points from 10 to 0 across each sprint, indicating consistent progress and timely completion of tasks. Each sprint maintains a balanced workload with high-priority features delivered first (validation, approval, notifications), followed by enhancements and integrations.